APSCO Cubesat Competition

Target Spacecraft Pose Estimation using Artificial Intelligence (AI) Methods in a 3U CubeSat

Conceptual and Preliminary Design Report

NSU-AI-SAT

North South University

Bangladesh

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1. Mission and Payload

Target Spacecraft Pose Estimation using Artificial Intelligence (AI) Methods in a 3U Cubesat

1.1. Mission

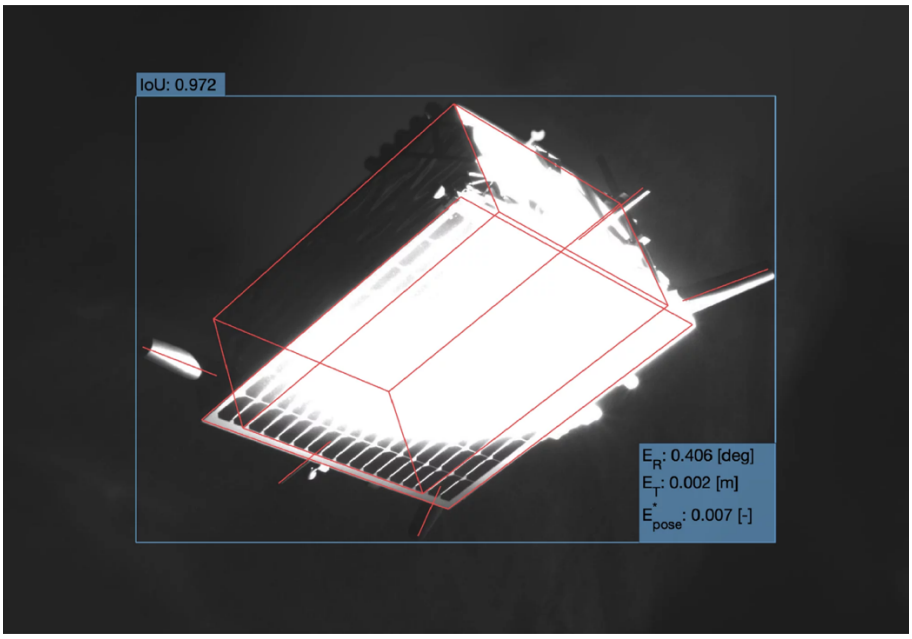

Figure 1- Satellite Pose Estimation Using Deep Learning [Reference provided in ACC mission proposal document]

The increasing congestion of Earth's orbital environment poses significant challenges to both existing satellite systems and future space missions. The prevalence of Space Resident Objects (SROs), which include cooperative and noncooperative satellites as well as space debris from previous missions, necessitates a comprehensive and technologically advanced handling approach. For future on-orbit servicing and debris removal missions, onboard pose estimation of a target spacecraft is an essential requirement.

To address this pressing issue, we propose **a 3U CubeSat mission for estimating the pose, the relative attitude and position, of a target spacecraft using a camera with state-of-the-art onboard computer vision and deep learning techniques**. In particular for this mission, we propose to proceed along the path of Artificial Intelligence (AI) based algorithm development proposed in the research literature.

Currently there is considerable interest in applying AI and deep learning methods in Satellite systems. For example, Figure 2 below shows pose estimation of a target spacecraft using convolutional neural networks [References are provided in the ACC mission proposal document].

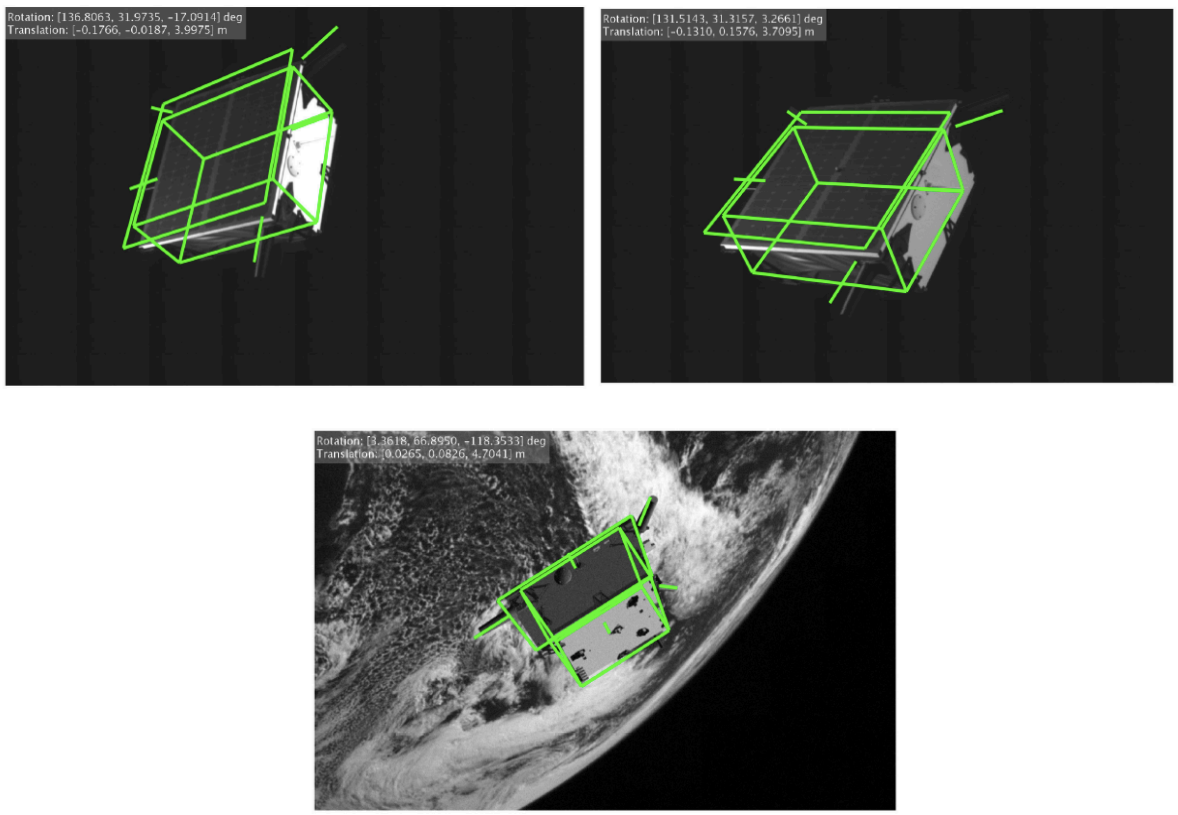

Figure 2- Satellite Pose Estimation Using Deep Learning [Reference provided in ACC mission proposal document]

A detailed description of the mission is provided in the ACC mission proposal document. Here we have updated the mission and provide the key requirements for the updated mission. Based on our conceptual and preliminary design study, some of the requirements have been changed from the ACC mission proposal document. The updated mission requirements are provided in Table 1.

Key mission requirements (updated from ACC mission proposal document)

Table 1 Updated key mission requirements

Key Parameter	**Value/Range**
Orbital Altitude	389.1 km (**Tiangong space station**)
Inclination	41.47 degrees
Orbital Period	92.3 minutes
Orbit Type	LEO
Camera	NanoCam C1U, updated from mission proposal document, previously a Monocular camera
Camera Resolution	2048 x 1536 (updated from the ACC mission proposal document)
Distance to Target Spacecraft	1 – 20 km, updated based on preliminary orbit simulation and camera specs (previously 7 – 30 m)
Camera Focal length	70.5 mm
Deep Learning Processing Unit (DPU)	Zynq Ultrascale+ MPSoC
DPU Power Consumption	7W/20W (idle/maximum)

In Table 1, based on preliminary orbit simulations, we have updated both the imaging distance to the target spacecraft and the camera model. The NSU-AI-SAT's orbit has to carefully designed to achieve proximity to orbiting space stations. Assuming NSU-AI-SAT will reach within 1 – 20 km of orbiting space stations, we have updated the camera to NanoCam C1U. At these distances the NanoCam C1U should be able to capture sufficient features of orbiting spacecraft for accurate pose estimation.

5. ADCS Attitude determination and control

In this section, we present the specifications for the Attitude Determination and Control System (ADCS) as they align with the Conceptual and Preliminary design phases of our project. This includes a detailed overview of the ADCS architecture, represented through a comprehensive block diagram that illustrates the integration of sensors, actuators, and processing units. Additionally, we delve into the operating features and parameters of the ADCS, detailing the accuracy, stability, sensor and actuator sizing, and various operational modes that are critical to meeting mission objectives. A budget estimation is also provided, which covers the mass, dimensions, power requirements, and costs associated with this subsystem. This structured approach ensures a clear understanding of the ADCS's capabilities and its role in the success of the mission.

1. Architecture

The Attitude Determination and Control Subsystem consists of sensors, actuators and an operating system. The module iADCS200 from AAC Clyde Space and Berlin Space Technologies has been chosen as the main board for the subsystem. It is a board with PC-104 compatible footprint. This fully autonomous integrated board consists of different built-in sensor/actuator components: the ST200 star tracker, with RW210 series reaction wheels and MTQ200 series magnetorquers with built-in IMU sensor.

For the mission specifications, the ADCS subsystem also required external sensor components. A total of 5 external analog Sun-sensors have been chosen, for which the module chosen is CubeSat Shop's nanoSSOC-A60. For magnetometers, Clyde Space's MM200 module and for Earth Horizon sensors, CubeSpace's module Gen2-CubeSense Earth have been selected.

The block diagram shown below describes the ADCS subsystem in a product tree down to the components.

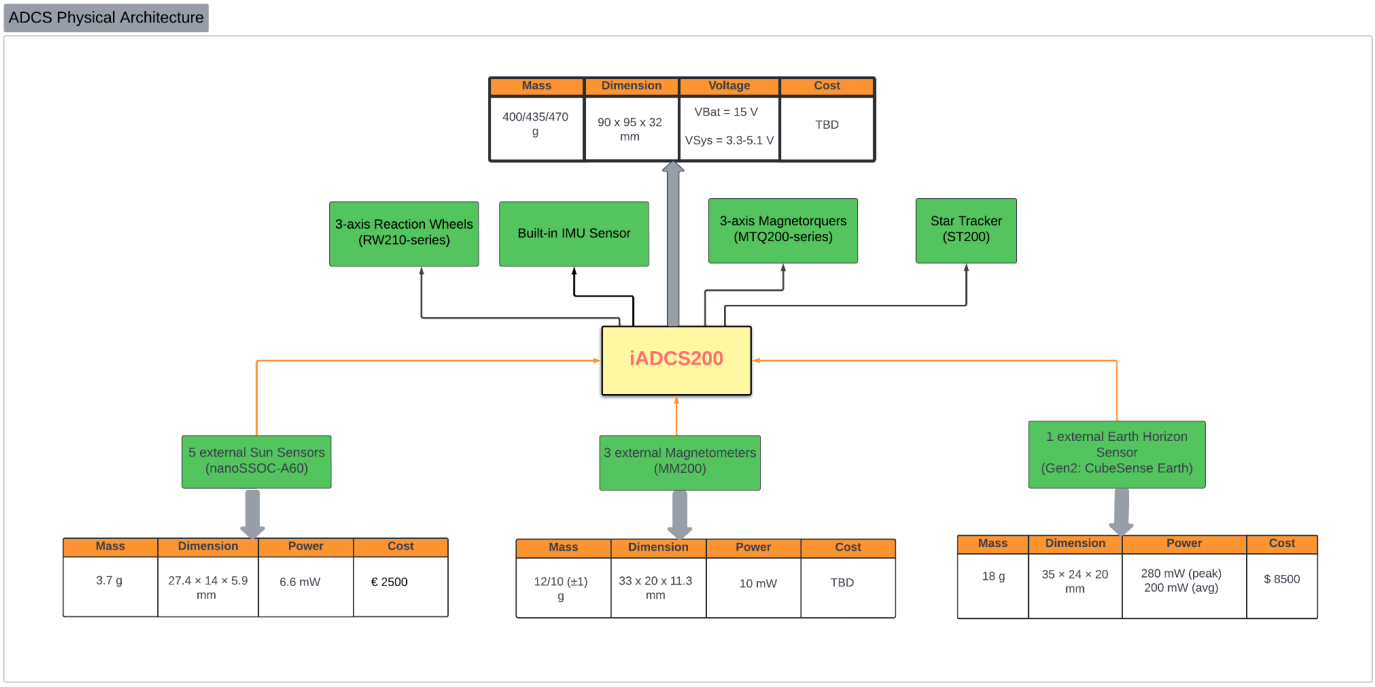

Figure 44 *Physical Architecture of ADCS*

2. Operating Features and Parameters

2.1 General operating parameters

For the analysis of the general operating parameters of the CubeSat, the first thing required is the axis definition for Attitude Determination or Control functionalities. The body frame axes are shown in the figure 45 -

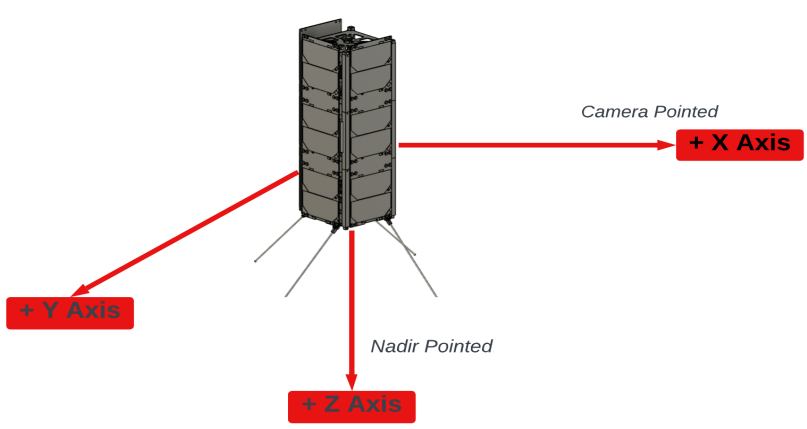

Figure 45 Body Frame Axis

The +X axis is in the orbital trajectory direction, i.e. the direction of the orbital path of the CubeSat. This axis is determined to be "Camera-Pointed". The +Z axis is in the Nadir direction, which is required to point the S-band antennas for communication with the Ground. The +Y axis is orthogonal to the +X and +Z axis.

The Mass is estimated to be 4 Kgs, Centre of Mass is considered to be at the geometric center and the principal inertia is taken as I = [Ixx Iyy Izz] = [0.033 0.033 0.00067]

2.1.1 Overview

The prime ADCS requirements have been summarized in the table below -

Table 16 *Summary of Requirements*

Accuracy Requirement	<1 deg/axis
Stability Requirement	<1 deg/s
Slewing Requirement	(22.5+-4)deg/s to (45+-4)deg/s

2.1.2 Nadir Pointing for Communications

This mode is required for the Satellite to establish Comms linkup with the ground station. And the system level requirement for pointing accuracy is estimated to be less than 1 deg along the S-band antenna pointed at the nadir (+Z axis) when the CubeSat is close to the ground station for data downlink/uplink. The required stability is estimated to be less than 1 degree per second, estimated to have a deviation of 2σ so that the ground station can maintain a clear communication with the CubeSat.

2.1.3 Sun Pointing for Solar Charging & Thermal Control

For optimal charging and passive thermal control, the CubeSat will have a basic sun pointing mode. That is, the +Y axis of the body frame is made to point towards the sun to achieve maximum incidence of sun rays; point any surface requiring thermal control to/ away from the incident sun rays.

2.1.4 Target Pointing for Imaging:

As the highest constraint is by the Target Pointing mode, we derive the justification for its implementation. A close approximation of the TLE data for the target is assumed to be known, however current Attitude & Angular rates of the target are unknown, which is our mission to make known. Our

CubeSat shall encounter our targets in passing orbits at multiple intervals as shown in Chapter 1.3 Orbital Analysis. This gives us a maximum range of 20 km, a minimum range of 1 km for Target Pointing.

A FOV (Field of View) region is constructed using the AER data and the Camera hardware constraints.

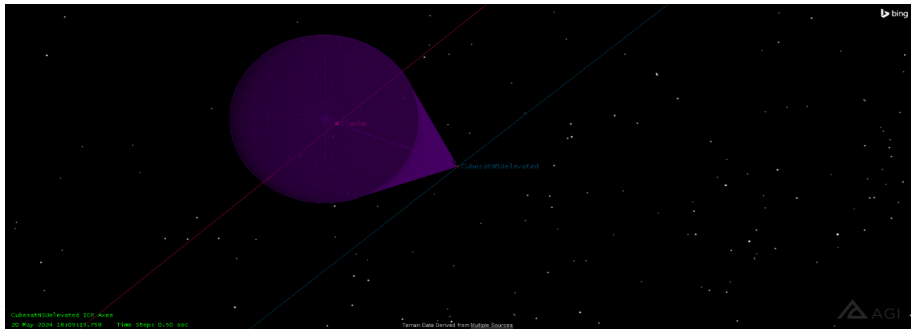

Figure 46 NSU AI SAT to Tianhe FOV

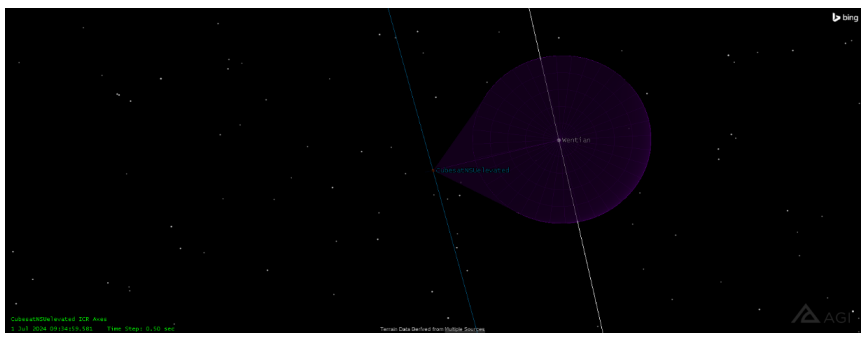

Figure 47 NSU AI SAT to Wentian FOV

For our selected payload, we obtain a IFOV (Inst. Field of View) of 0.003 degrees. The ground sampling distance is 0.9231 m/pixel over 20KM and 0.09 m/pixel over 2KM. The target satellites are all under 20km, giving us an approximate target swath of 20 by 20 pixels over 200km and 223 by 223 pixels apprx. over 2KM. Thus, the sharpest possible image can be achieved with a pointing stability of 0.003 degree per axis, however, with AI implementation, sub-par visibility is also acceptable. So, we aim for <<1(1σ) deg pointing accuracy with <1 deg/s stability along each half cone axis

After the Line of Sight is established, the ADCS shall aim to keep the target within the FOV with the camera face of the body frame pointing towards the target. Considering a max slew deviation of 0.0698 rad/s(4 deg/s) of the CubeSat itself, [iadcs200 rated for 0.026529 rad/s (1.52deg/sec)] and the minimum duration of close approach to be 4 seconds, the CubeSat must be able to perform a minimum slew

maneuver of (90+-4) degrees to a maximum of (180+-4) degrees under 4 seconds, that is, (22.5+-4)deg/s to (45+-4)deg/s depending on initial attitude on approach.

In Summary,

Table 17 Target pointing accuracy and stability requirement

Modes Requiring Pointing	Pointing Accuracy	Pointing Stability
Nadir Pointing for Communications	<<1 deg/axis (1σ)	<1 deg/s
Sun Tracking for Maximal Charging & Thermal Control	<5deg/axis	-
Target Pointing for Imaging	<1 deg/axis (3σ)	<1 deg/s

2.2 Sensors sizing and Actuators sizing

2.2.1 Approach

For Sensor Sizing & Actuator Sizing, we use a combination of the Nanostar Methodology and an existing structured simulation environment. The MATLAB simulation is adapted to the Orbit Parameters and initialized with the STK generated TLE file for the NSU CubeSAT - "CNSU.TLE". By testing the propagated orbit & matching it with STK output, we can ascertain that the simulation is functional for base parameters.

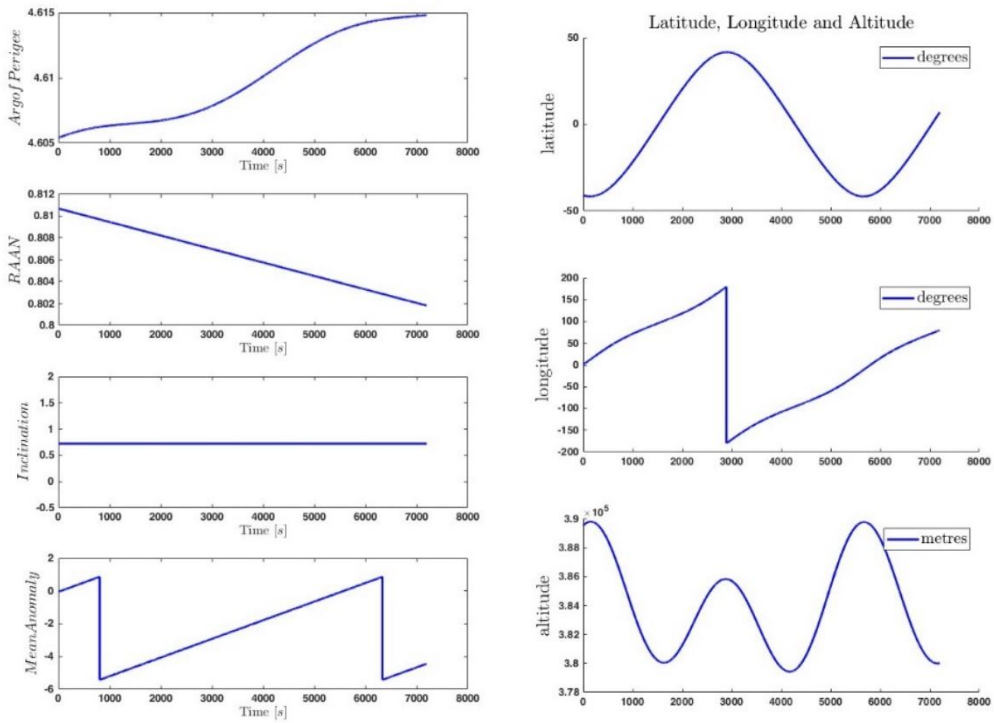

Figure 48 MATLAB Orbit Parameters

2.2.2 Disturbance Modelling[8]

A 3U CubeSat in LEO orbit can have primarily 4 external disturbances and some negligible internal disturbances for simplified application, expressed in torques.

Table 18 Disturbance Torque Estimation

Disturbance Torques	Mathematical Model	Calculated Max Values
Gravity Gradient Torque	$T_g = \frac{3\mu}{{r_p}^3} * \lvert I_{max} - I_{min} \rvert * sin(2 * \theta)$	"45.3912" $\times 10^{-9}$

[8] Wertz, J.R., et al. (2011). Space mission engineering: The new SMAD. Hawthorne, CA, Microcosm Press

Solar Radiation Pressure (SRP) Torque	$T_{sp} = \frac{\Phi}{c_{light}}.A_{sp}.(1+q).cos(i_{suninc}).\|\|c_{sp} - c_g\|\|$	"21.1202" $\times 10^{-9}$
Aerodynamic Torque	$T_a = \frac{1}{2}.\rho_{atm,p}.C_D.A_a.V_p{}^2.\left\|\left\|c_a - c_g\right\|\right\|$	"2.01615" $\times 10^{-6}$
Magnetic Torque	$T_m \approx D_m.(\frac{\lambda.M_m}{r_p{}^3})$	"1.80293" $\times 10^{-5}$
Negligible Torques Due to Internal Disturbances	$T_{miscellaneous}$	0
Total	$T_D = T_g + T_{sp} + T_a + T_m + T_{misc}$	"2.01119" $\times 10^{-5}$

The disturbances are modeled mathematically using the referred equations with values from models like- NRLMSISE 2, IGRF and worst-case scenarios at 390 km altitude.

2.2.3 Actuator Sizing

2.2.3.1 Constraints Identification

Matlab Simulation for the detumbling mode is run through trial and error to identify the minimal MTQ torque required to stabilize the CubeSat in detumbling mode and the minimum number of necessary orbits. It was found that for maximum torque greater than 0.05 A.Nm, the CubeSat can be successfully stabilized within 8 orbits (12 Hours).

For nominal mode, the Matlab Simulation is run through trial and error to identify the minimal MTQ & RW torque required to stabilize the CubeSat in the same time frame. By iterating the simulation for various combinations of MTQ's max applied torque, RW's max applied torque, angular velocity, and reaction wheel moment of inertia, the minimum possible values (Quaternion & Angular Velocity) are calculated for which stability can be achieved.

Now, using the NanoStar Methodology for max required torques, the maximum for the same parameters is calculated, both of which supersede the maximum values derived from the simulation.

Table 19 Torque & Momentum Budget Calculation

Calculated Maximum Required Values			
Magnetorquers		**Reaction Wheels**	
Maximum Magnetic Dipole Moment, $D_{m,MT} = T_D.(\frac{{r_p}^3}{2.M_m})$	"0.39043"	*For Disturbances,* $T_{RW,d} = T_D.(1+SM)$	"2.41343" $\times 10^{-5}$
		For Slew Maneuver, $T_{RW,s} = 4.thet2.(\frac{\pi}{180}).(\frac{I_s}{(60.t_s)^2})$	"1.11265" $\times 10^{-5}$
		Max Torque for RW, $T_{RW} = max\{T_{RW,d}, T_{RW,s}\}$	"2.41343" $\times 10^{-5}$
		Max Stored Angular Momentum, $H_{RW} = T_D.P$	0.111259

2.2.3.2 Magnetorquer Selection

For the MTQ, the derived constraints are shown in the table below -

Table 20 MTQ Constraints Estimation

-	**Dipole Moment (A.m2)**	
-	Minimum Required for Detumbling	Minimum Required for Nominal
Simulated	0.05	0.20
Calculated	-	**0.37**

While the required dipole moment constraint for nominal is 0.37, it is found in the simulations that desirable stabilization can be achieved both in detumbling as well as other pointing modes at nearly half the dipole moment with the joint operation of Reaction wheels, albeit within a longer, but acceptable duration for our mission.

The module MTQ200.20 is thus selected for our use-case with the performance of:

Table 21 MTQ200.20 Performance Parameters

Nominal Dipole Moment	0.2	A.m2
Boost Dipole Moment	Upto 1	A.m2
Control Accuracy	1	mA.m2 steps

2.2.3.3 Reaction Wheel Selection

For the RW, the derived constraints are shown in the tables below -

Table 22 RW Constraints Estimation-A

-	**Torque (N.m)**		**Axial Moment of Inertia (kgm2)**	
-	Minimum with Higher RPMS	Minimum with Sufficient RPMs	Minimum with Higher RPMs	Minimum with Sufficient RPMs
Simulated	5e-6	1e-4	5e-7	1.9e-7
Calculated	-	**2.291e-5**	-	-

Table 23 RW Constraints Estimation-B

-	**Angular Velocity (rpm)**		**Momentum Storage Per Axis**	
-	Minimum Required	Maximum Required	Minimum Required	Maximum Required
Simulated	9000	14000	-	-
Calculated	-	-	-	0.105609

The RW210-v3 by AAC Clyde Space is selected with the performance:

Table 24 RW210v3 Performance Parameters

Total momentum storage	+/- 1.5, +/- 3.0, +/- 6.0	mN.m.s
Maximum torque	+/- 0.1	mN.m
Maximum rotation rate	10000 / 15000 / 15000	rpm
Control accuracy	+/- 0.5	rpm

2.2.4 Sensor Sizing

As denoted in requirements, our mission parameters demand great stability, primarily for camera pointing, so high-precision sensors are selected within budget to keep noise levels to the absolute minimum possible.

2.2.4.1 Magnetometer

For our CubeSat, we require a magnetometer to provide the attitude with respect to the local magnetic field of the Earth. A 3-axis magnetometer providing readings for each of the 3-axis of our body frame with good precision and low noise is sufficient. The primary noise sources here are the residual dipole moment of the CubeSat and the estimation error of the local magnetic field. We select MM200 by AAC Clyde Space for our use-case, providing the following performance:

Table 25 MM200 Performance Parameters

Sampling rate	Max 500	Hz (configurable)
Range	+800	µT
Total noise spectral density	1.18	nT/√Hz

2.2.4.2 Sun Sensor

5 sun sensors of the same model are utilized to provide an accurate attitude determination. For our system modeling, we define 5 frames, one for every sensor. Earth Albedo Effect is the primary noise source for the sun sensors; the Albedo radiation is assumed to be coming from the nadir direction. We select nanoSSOC-A60 for our use-case, providing the following performance:

Table 26 NanoSSOC-A60 Performance Parameters

Sensor type	2 axes	Orthogonal
Field of view (FOV)	± 60 º	Performance Field
Accuracy	< 0.5 º	3σ error
Precision	< 0.1º	

2.2.4.3 Built-In IMU (Gyroscope)

The module iadcs200 comes built-in IMU. Considering the high precision already provided by the datasheet, it is inferred that it suffices for our use case.

Table 27 *iADCS200 IMU Performance Parameters*

Attitude determination accuracy	30	arcseconds (3-sigma)

2.2.4.4 Earth Horizon Sensor

It is evident from our simulations and literature review that an ADCS based only on Sun Sensors, Magnetometers & Gyro performs poorly in eclipse periods. To avoid repeated detumbling and image acquisition even during the eclipse period, an Earth Horizon Sensor is integrated. The CubeSense Earth Gen 2 is selected for our use-case with the performance:

Table 28 *CubeSense Earth Gen 2 Performance Parameters*

Accuracy [3-sigma]	1° (roll and elevation)

2.2.4.5 Star Tracker

To further compensate for eclipse accuracy and overall attitude determination, a Star Tracker is also integrated.

The ST200 built in Star Tracker with the iadcs200 by AAC Clyde Space provides the following performance:

Table 29 ST200 Performance Parameters

Attitude determination accuracy (pitch, yaw)	30	arcseconds (3-sigma)
Attitude determination accuracy (roll)	200	arcseconds (3-sigma)
Maximum slew rate (tip/tilt)	< 0.3	°/s
Maximum slew rate (roll)	< 0.6	°/s

2.2.5 Simulation

The MATLAB simulation is performed over 1 orbital period by updating the sensor information and noise estimation with our estimated disturbances, selected actuators, and selected sensors in the modeling methodology here.[ref] A B-dot Controller is implemented for Detumbling, a PD controller for Nominal with detumbling switching and a MEKF filter for attitude determination from multiple sensors.

We use the ECS error-index to ascertain that, indeed, optimal pointing (here, towards q_desired = [1 0 0 0]) is achievable with our selected combination of actuators and sensors in our estimated mission environment.

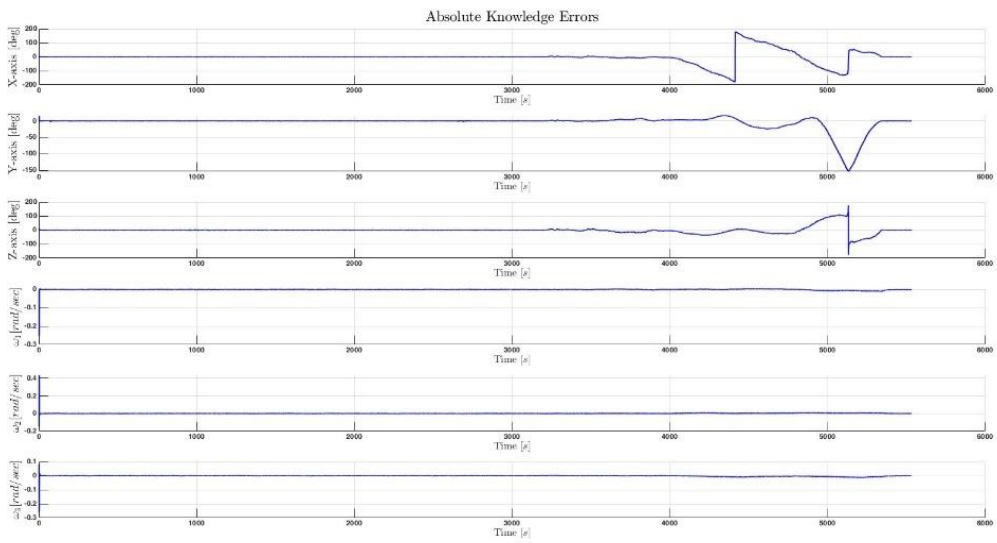

Figure 49 ***Absolute Knowledge Error over 1 Orbit***

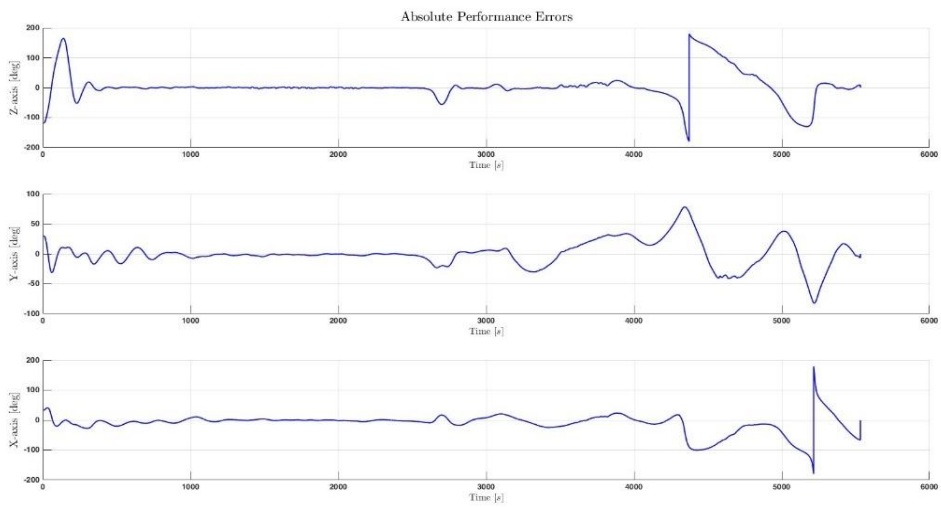

Figure 50 ***Absolute Performance Error over 1 Orbit***

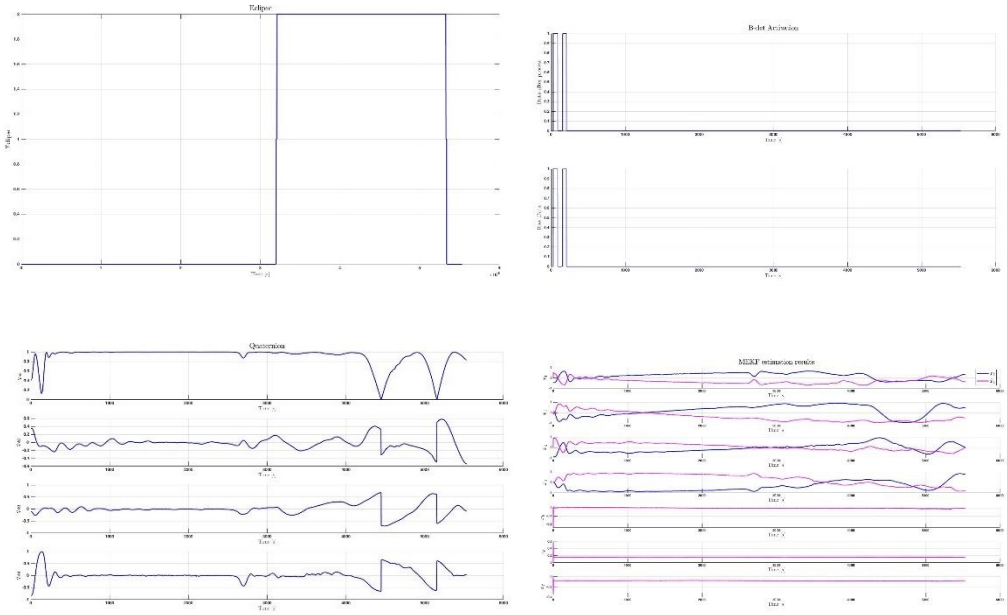

Figure 51 ***[TL]:Eclipse Period; [TR]: B-dot Activation; [BL]: Quaternions; [BR]: MEKF Est***

Again, in solely Detumbling scenario as well, we achieve sufficient stabilization -

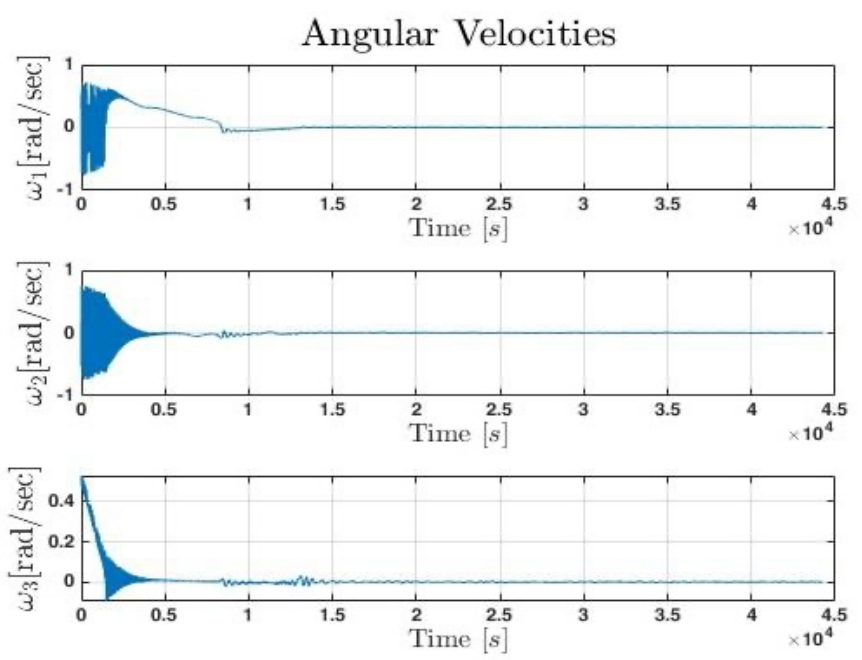

Figure 52 ***Bdot Control over 8 orbits with only MTQ where mtq_max=0.2.***

2.3 Operation Modes

In this section, each mode of ADCS is outlined. The functional architecture of the ADCS is provided in Fig. 53.

a. Launch/Deployment:

This is the first stage of a satellite mission. At this stage, the CubeSat is released from the P-POD of the spacecraft for which the Attitude actuation and determination will not be necessary. Only the attitude sensor measurements and data collection function will be active.

b. Detumbling:

This is the next stage of the satellite mission after launch/deployment. This stage is where the first control strategy or maneuver is applied. Usually after launch the satellite moves at a very high angular velocity rate. For the mission, the minimum angular rate is established as 5 deg/s. So, if the angular rate of the satellite is greater than 5 deg/s, the detumbling mode turns on. The B-Dot algorithm is used as the control strategy in this mode. With the help of the 3-axis Magnetorquer, the angular velocity of the satellite is saturated. By exerting a magnetic torque against the magnetic field of the earth so that the magnetorquer stabilizes the motion/spin of the satellite as per requirement. If the dipole moment exceeds 0.20 A-m^2 the Reaction Wheel turns on to assist the magnetorquer for satellite stabilization. The detumbling mode ends when the CubeSat is stabilized so that its angular velocity becomes less than 1 deg/s.

c. Nadir Pointing:

As soon as the detumbling mode ends, the CubeSat needs to be Nadir-pointed, or at least the antenna of the CubeSat needs to be pointed in the nadir direction. Just after the CubeSat stabilizes from the tumbling, it needs to communicate with the ground station (GS). For this reason, the antenna needs to be in the nadir direction. For nadir pointing, a simple PID controller can be used to maintain a skew maneuvering. Reaction wheel is used in this mode.

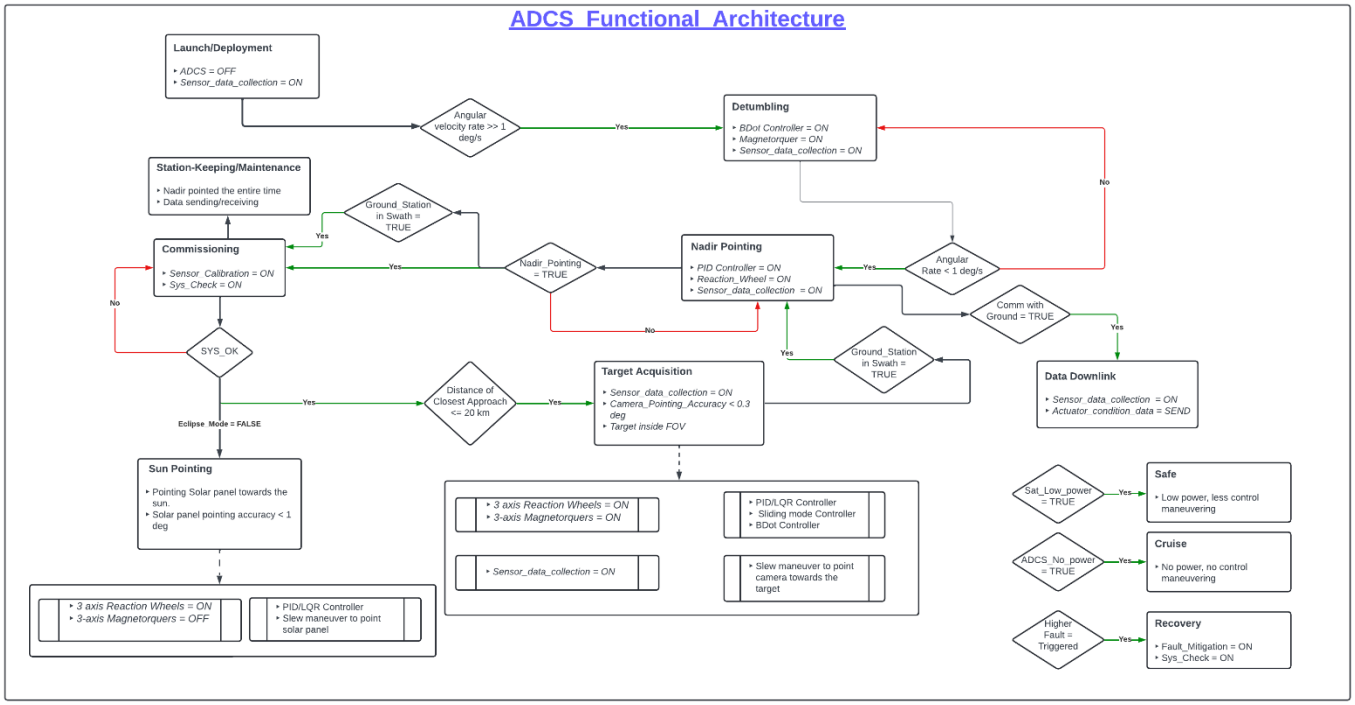

Figure 53 ***Functional Architecture of ADCS***

d. Commissioning:

Once the swath of the CubeSat contains the ground station, the commissioning mode begins. At this stage, the ADCS sensors and actuators are calibrated. The sensor data collection is turned on to transmit the house-keeping data to the ground station.

e. Station-Keeping/Maintenance:

When the ground station is in the swath area of the CubeSat, the station-keeping/maintenance takes place. The ground station communicates with the CubeSat at this stage and sends/receives the telemetry command and data respectively. For this mode, the satellite needs to be stable in the nadir direction, so that a proper communication is maintained.

f. Data Downlink:

Usually at the station-keeping/maintenance stage, the data downlink mode starts. For this stage/mode, the main objective is to establish a proper communication with the ground station. Despite the efficiency of the Communication or Telemetry Handling subsystem, for a successful data handling mission a stable ADCS control is required. For this, the Reaction wheel is controlled to keep stability. At

this stage, the sensor and actuator data collection functions are on, so that the most current data can be sent to ground control.

g. Sun Pointing:

Except for the Eclipse portion of the mission, the ADCS will control the CubeSat to direct the solar panels towards the sun. This direction can be obtained from the sun-vector measured from the sun-sensors and determined from the filtering and calculations onboard. The required pointing accuracy for the solar panels is less than 1 deg. This can be done with the help of the 3-axis Reaction momentum wheels. A PID or LQR controller can be used for adjusting the CubeSat according to the requirement.

h. Target Acquisition:

This stage is where the prime mission of the CubeSat takes place. When the distance of the closest approach value of CubeSat from the target spacecraft is equal to 2 km (2000 m), the Target Acquisition mode starts. For this mode, the camera needs to be pointed at the target spacecraft and the pointing accuracy needs to be less than 0.3 deg. Another requirement of this mode is that the Target needs to be in the FOV (Field of View) of the payload camera. For the entire time till the images have been captured, the CubeSat needs to be stabilized.

Since this stage is a very crucial stage of the Satellite modes, the CubeSat needs to be at the most stable form. For this, all the necessary controllers need to be active. From a hardware perspective, the Reaction Wheels and Magnetorquers need to be active. For necessary slew maneuver, the required axis needs to be rotated with the actuators.

i. Cruise:

The Cruise mode is on when the CubeSat ADCS is not required in the certain time period of the satellite mission. This mode is considered as an idle state for the ADCS components, to restore energy for the actuators.

j. Safe:

The Safe mode is on when the CubeSat ADCS obtains a lower-level fault, like a Warning. At this stage, the ADCS will function with low power.

k. Recovery:

The CubeSat reaches this mode only when a higher-level fault is triggered, for example – if the sensors or actuators stop communicating. The ADCS subsystem starts functioning again in different modes only when the Fault has been mitigated. At this stage, the system check is always on, till the fault is untriggered.

3. Budget Estimation

The ADCS budget estimation has been showcased in the following table:

Table 30 ADCS Budget Estimation

	Components	Module	Number of Components	Mass (g)	Dimension (mm)	Power (mW)	Cost
iADCS200	Reaction Wheel (RW210 Series)	RW210 Series	1	470	95 x 90 x 32	4000 (peak)	N/A
	Magnetorquer	MTQ200 Series	3			1400 (Nominal)	
	Star Tracker	ST200	1			1150 (Idle)	
External Components	Sun sensors	nanoSSOC-A60	5	18.5	27.4 x 14 x 5.9 (each)	6.6 (peak)	£ 2500
	Magnetometer	MM200	3	36	33 x 20 x 11.3 (each)	10 (peak)	N/A
	Earth Horizon Sensor	Gen2-Cubesense Earth	1	18	35 x 24 x 20	280 (peak)	$8500

10. Sum up

At the conclusion of the conceptual and preliminary design stage, we gained a number of insights into NSU-AI-SAT. These lessons learned are summarized below.

1. Orbit design is an extremely important component of NSU-AI-SAT's mission. To achieve good performance in spacecraft pose estimation it is essential to capture good quality images which is possible by minimizing the distance with the target spacecraft. With a carefully designed orbit, we will be able to bring NSU-AI-SAT in close proximity to a number of orbiting space stations, providing the opportunity to capture high quality images with sufficient features for high-performance pose estimation.

A carefully designed orbit will allow us to increase the number of closest approach events with orbiting space stations, decrease the closest approach distance and increase the closest approach duration. In the next stage, we plan to thoroughly investigate the orbit design problem for NSU-AI-SAT.

2. At the end of this stage, we learned that NSU-AI-SAT has 0.55U of empty space and a mass of 2.99 kg. With a careful rearrangement of the components, we can easily accommodate a larger battery pack in NSU-AI-SAT. In the next stage we would like to incorporate the GOMspace NanoPower BPX battery in NSU-AI-SAT.

https://gomspace.com/shop/subsystems/power/nanopower-bpx.aspx

The GOMspace NanoPower BPX battery would add 0.18U volume and 230 g extra mass to NSU-AI-SAT but would be a high-performance power source.

In the next stage, we plan to act upon these lessons learned.